

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Eksamen

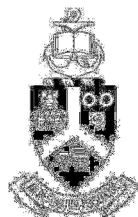
Kopiereg voorbehou

Examination

Copyright reserved

Module EBN122
Elektrisiteit en Elektronika
23 November 2015

Module EBN122
Electricity and Electronics
23 November 2015



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting: Exam information:

Maksimum punte: Maximum marks:	91	Volpunte: Full marks:	90
Duur van vraestel: Duration of paper:	180 minute 180 minutes	Oopboek / toeboek: Open / closed book:	Toe Closed
Totale aantal bladsye (hierdie blad ingesluit): Total number of pages (including this page):		25	

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
- Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Polêre vorm: $A \angle \theta$; Reghoekige vorm: $a + jb$
Polar form: $A \angle \theta$; Rectangular form: $a + jb$
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Dosent(e): Lecturer(s):	J. Joubert
	A. Chatterjee
Eksterne eksaminator: External examiner:	J.W. Odendaal

VRAAG/QUESTION 1 [45]

Vraag 1.1 [1]

Hoeveel elektrone is nodig om 'n lading van $20pC$ te produseer.

Question 1.1 [1]

How many electrons are required to produce a charge of $20pC$.

Vraag 1.2 [2]

Bepaal die stroom deur 'n element by $t = 0.2s$, as die lading gegee word deur $q(t) = 0.1e^{6t} + 2t$ C.

Question 1.2 [2]

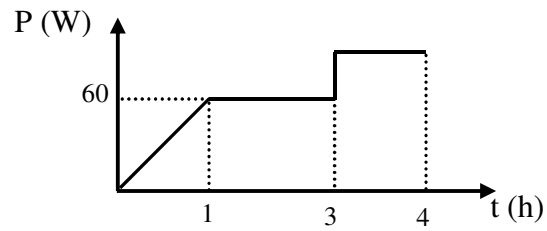
Find the current flowing through an element at $t = 0.2s$, if the charge is given by $q(t) = 0.1e^{6t} + 2t$ C.

Vraag 1.3 [2]

Die drywing wat gedissipeer word deur 'n komponent word getoon. Hoeveel energie is verkwis gedurende die eerste 2 ure. Gee jou antwoord in kJ .

Question 1.3 [2]

The power dissipated by a device is shown. Find the energy consumed by this device in the first 2 hours. Give your answer in kJ .

**Vraag 1.4 [2]**

'n Yskas is gespesifiseer as 300W. Bepaal die maandelikse koste (in Rand) om die yskas te gebruik vir 24 uur 'n dag, 30 dae 'n maand. Die koste van elektrisiteit is R0.9/kWh.

Question 1.4 [2]

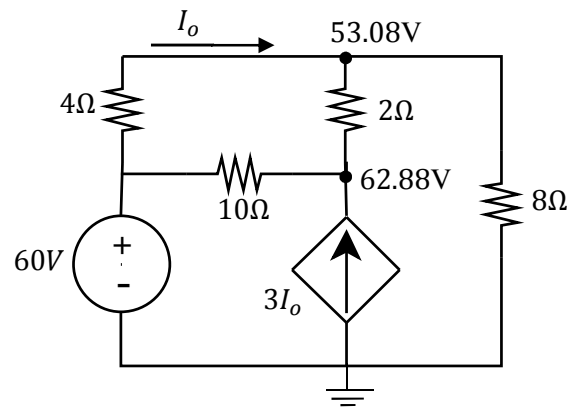
A fridge is rated at 300W. Find the monthly cost (in Rands) of running this fridge if it is used for 24 hours a day, 30 days a month. The cost of electricity is R0.9/kWh.

Vraag 1.5 [4]

- a) Bereken die drywing ge-assosieer met die afhanklike stroombron.
- b) Bereken die drywing ge-assosieer met die onafhanklike spanningsbron.

Question 1.5 [4]

- a) Find the power associated with the dependent current source.
- b) Find the power associated with the independent voltage source.



Vraag 1.6 [2]

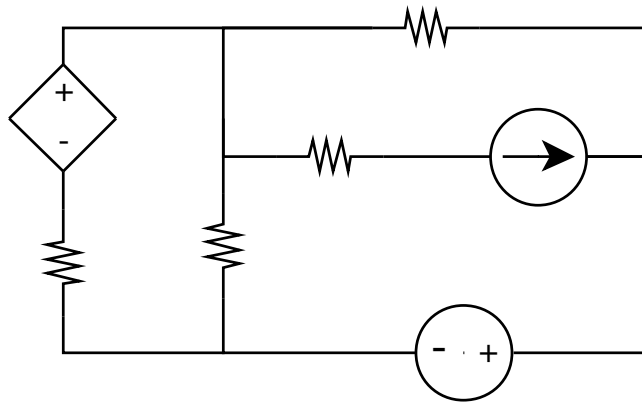
Bepaal:

- a) Die aantal nodes.
- b) Die aantal takke.

Question 1.6 [2]

Find:

- a) The number of nodes.
- b) The number of branches.



Vraag 1.7 [4]

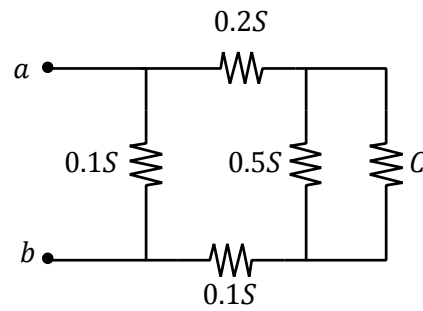
Bepaal die ekwivalente weerstand soos gesien deur terminale a-b as:

- a) $C = 0S$.
- b) $C = \infty S$.

Question 1.7 [4]

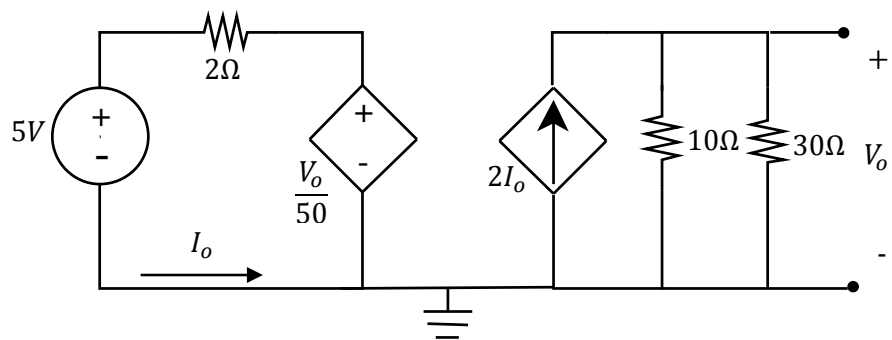
Find the equivalent resistance as seen by terminals a-b if:

- a) $C = 0S$.
- b) $C = \infty S$.



Vraag 1.8 [4]
Bepaal V_o en I_o .

Question 1.8 [4]
Find V_o and I_o .



Vraag 1.9 [4]

Gebruik node-analise om twee vergelykings van die volgende vorm op te stel:

$$aV_1 + bV_2 = 5$$

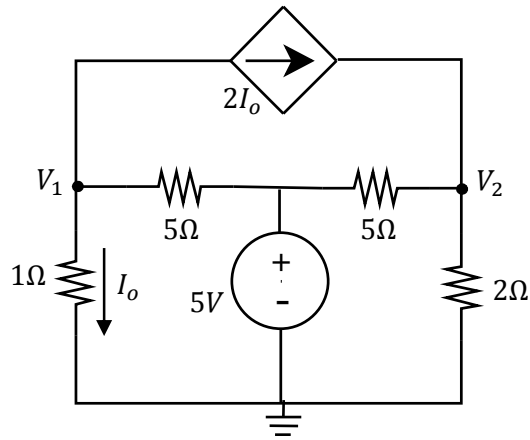
$$cV_1 + 7V_2 = d$$

Question 1.9 [4]

Use nodal analysis to set up two equations of the following form:

$$aV_1 + bV_2 = 5$$

$$cV_1 + 7V_2 = d$$



Vraag 1.10 [4]

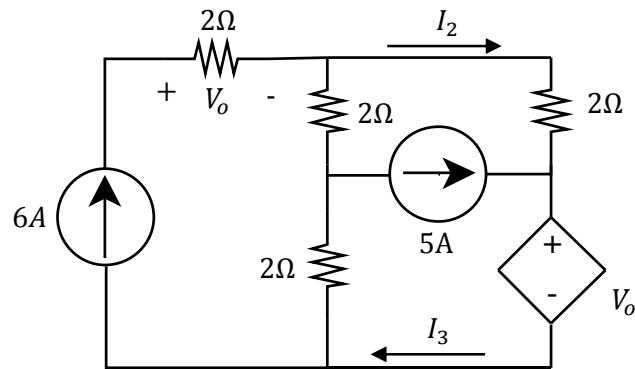
Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel:

$$eI_2 + fI_3 = 12 \quad (\text{KVL})$$
$$gI_2 - I_3 = h$$

Question 1.10 [4]

Use mesh analysis to set up two equations of the following form:

$$eI_2 + fI_3 = 12 \quad (\text{KVL})$$
$$gI_2 - I_3 = h$$



Vraag 1.11 [4]

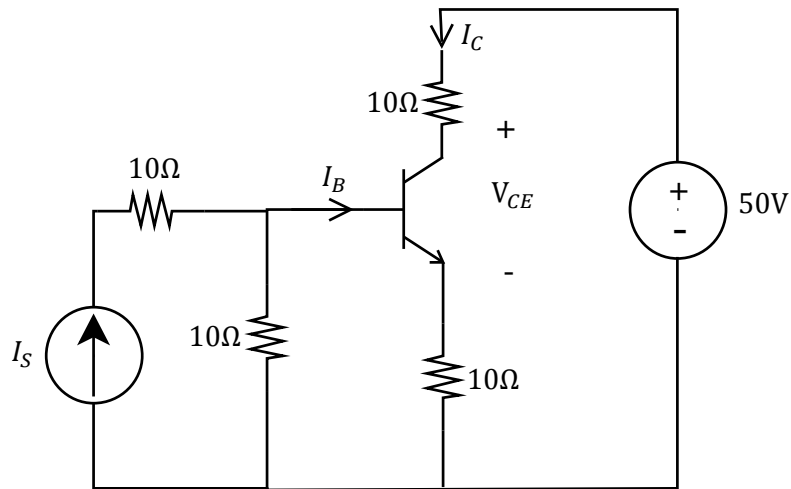
Aanvaar $V_{BE} = 0.7\text{V}$, en $\beta = 100$.

- a) Aanvaar $I_s = 3\text{A}$ en bepaal die basis-stroom I_B (in mA).
b) Aanvaar $I_C = 2\text{A}$. Bepaal V_{CE} . (Dit is nie nodig om I_s se waarde te weet nie).

Question 1.11 [4]

Assume $V_{BE} = 0.7\text{V}$, and $\beta = 100$.

- a) Assume $I_s = 3\text{A}$ and find the base current I_B (in mA).
b) Assume $I_C = 2\text{A}$. Find V_{CE} . (Knowledge of I_s is not required).



Vraag Question 1.12 [4]

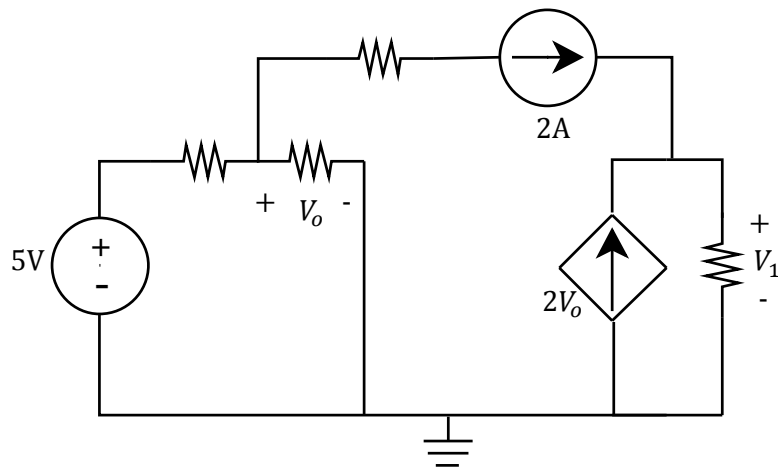
Alle weerstande is 10Ω .

- a) Bepaal V_I slegs agv die 2A stroombron.
- b) Bepaal V_I slegs agv die 5V spanningsbron.

Question 1.12 [4]

All resistor values are 10Ω .

- a) Find V_I due to the 2A current source only.
- b) Find V_I due to the 5V voltage source only.

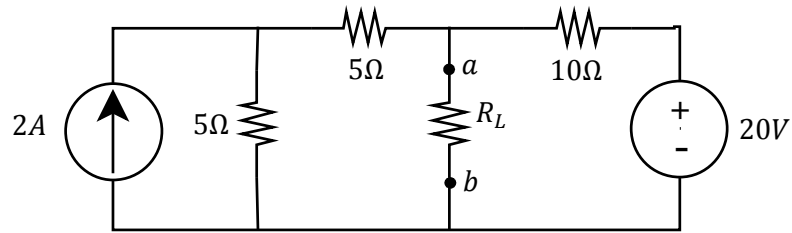


Vraag 1.13 [2]

- a) Bepaal die Norton ekwivalente stroom tov terminale a - b .
- b) Bepaal die waarde van die lasweerstand, R_L , sodat maksimum drywing oorgedra word.

Question 1.13 [4]

- a) Find the Norton equivalent current wrt terminals a - b .
- b) Find the value of the load resistor, R_L , such that maximum power is transferred to it.

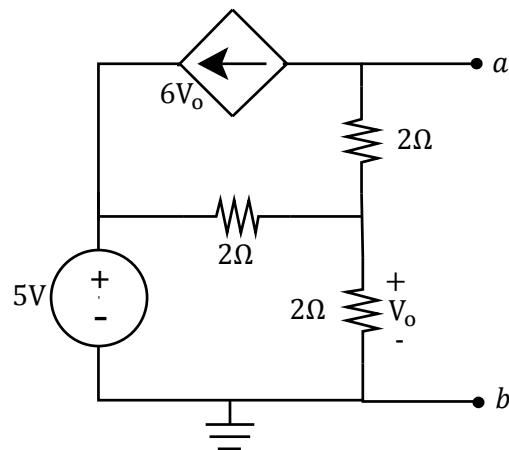


Vraag 1.14 [4]

Bepaal die Thevenin ekwivalente stroombaan ten opsigte van terminale a - b .

Question 1.14 [4]

Find the Thevenin equivalent circuit with respect to terminals a - b .



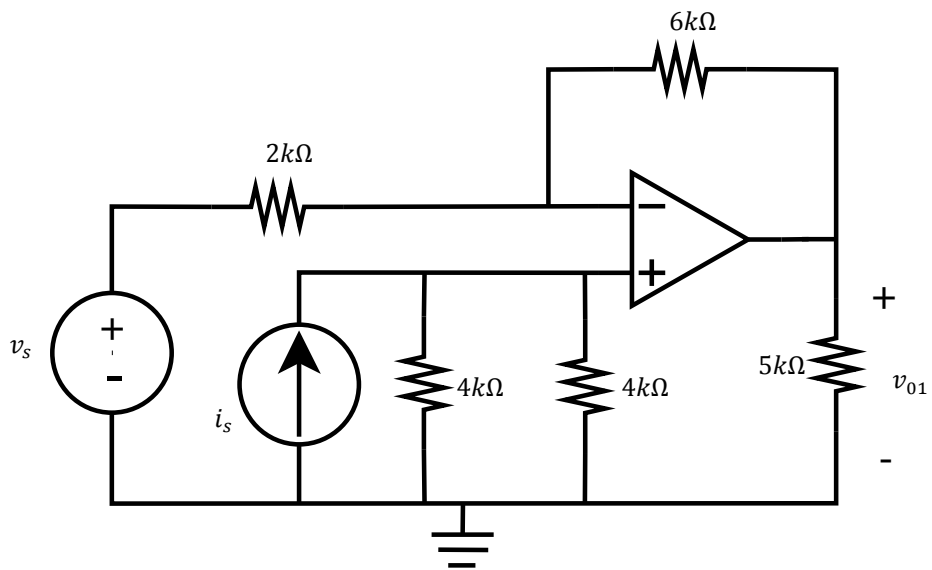
VRAAG/QUESTION 2 [22]

Vraag 2.1 [4]

- a) Bepaal v_{01} indien $v_s = 3V$ en $i_s = 1\text{ mA}$.
b) Bepaal v_{01} indien $v_s = -2V$ en $i_s = 0\text{ mA}$.

Question 2.1 [4]

- a) Determine v_{01} if $v_s = 3V$ and $i_s = 1\text{ mA}$.
b) Determine v_{01} if $v_s = -2V$ and $i_s = 0\text{ mA}$.

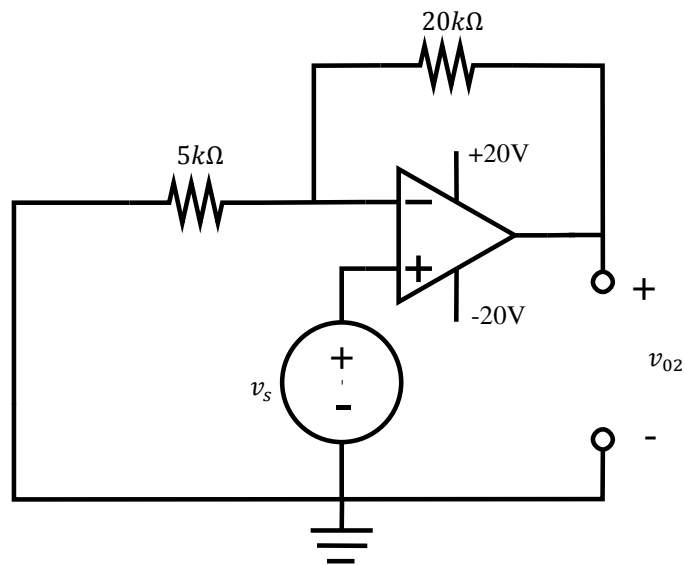


Vraag 2.2 [4]

- a) Bepaal v_{02} as $v_s = 2V$.
b) Wat is die maksimum waarde van v_s voordat die OV in versadiging sal ingaan.

Question 2.2 [4]

- a) Determine v_{02} if $v_s = 2V$.
b) What is the maximum value v_s can be before the op-amp goes into saturation.

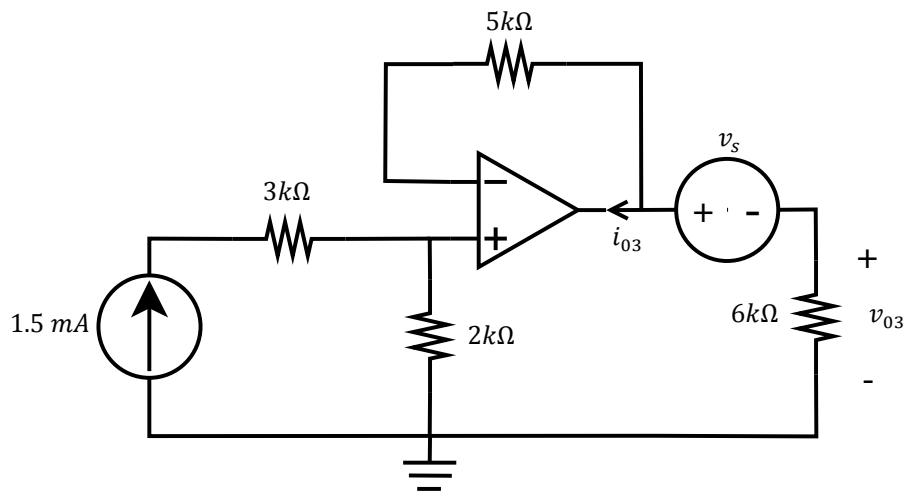


Vraag 2.3 [4]

- a) Bepaal v_{03} as $v_s = 3V$.
b) Vervang die stroombron met 'n 1.5V spanningsbron (met die + terminaal gekoppel aan die $3k\Omega$ weerstand). Bepaal i_{03} (in mA) as $v_s = 0V$.

Question 2.3 [4]

- a) Determine v_{03} if $v_s = 3V$.
b) Replace the current source with a 1.5V voltage source (with the + terminal connected to the $3k\Omega$ resistor). Determine i_{03} (in mA) if $v_s = 0V$.

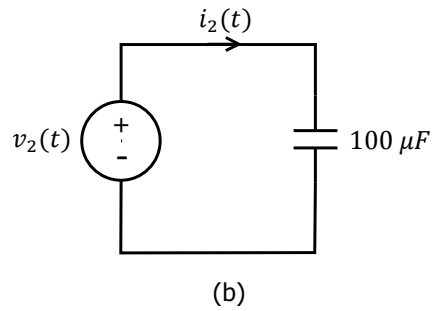
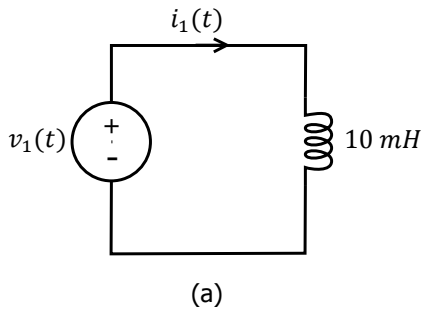


Vraag 2.4 [4]

- a) Die spanning $v_1(t) = 3 \cos(t + 30^\circ) V$ word aangelê oor 'n 10 mH induktor (sien Figuur (a)). Bepaal i_1 by $t = 2s$. (aanvaar $i_1(0) = 0$).
- b) Die spanning $v_2(t) = 3t^2 - 2t$ word aangelê oor 'n $100 \mu F$ kapasitor (sien Figuur (b)). Bepaal i_2 by $t = 4s$.

Question 2.4 [4]

- a) The voltage $v_1(t) = 3 \cos(t + 30^\circ) V$ is applied across a 10 mH inductor (see Figure (a)). Determine i_1 at $t = 2s$. (assume $i_1(0) = 0$).
- b) The voltage $v_2(t) = 3t^2 - 2t$ is applied across a $100 \mu F$ capacitor (see Figure (b)). Determine i_2 at $t = 4s$.

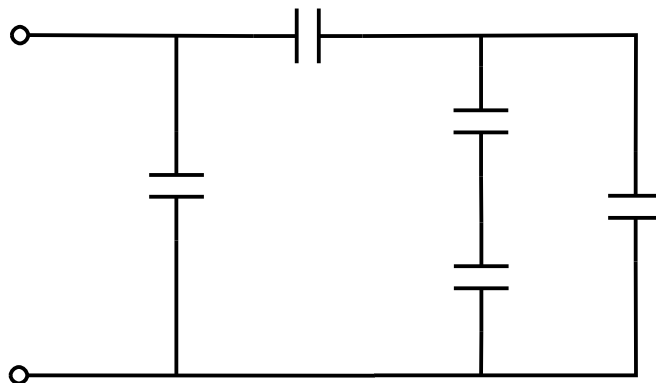


Vraag 2.5 [2]

Bepaal die ekwivalente kapasitansie – aanvaar alle kapasitore het 'n waarde van $10mF$.

Question 2.5 [2]

Determine the equivalent capacitance – assume all capacitors have a value of $10mF$.

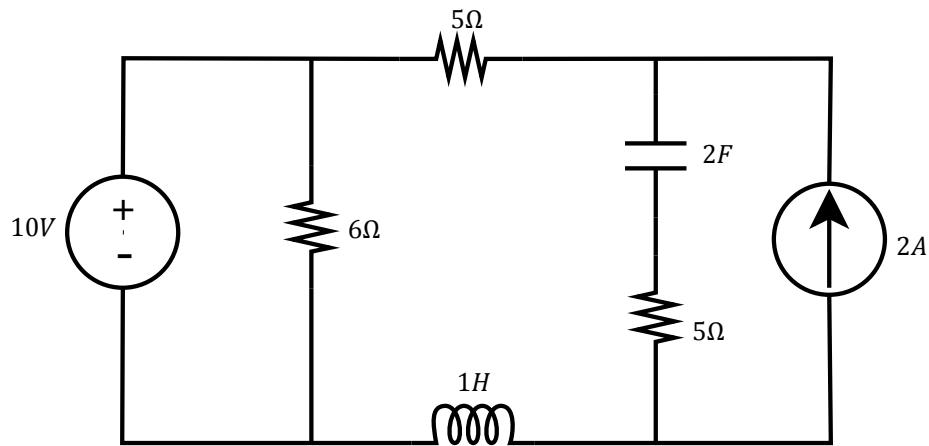


Vraag 2.6 [4]

Aanvaar die baan is in 'n stabiele toestand. Bepaal die gestoorde energie in die induktor en die kapasitor.

Question 2.6 [4]

Assume steady state conditions in the circuit. Determine the energy stored in the inductor and the capacitor.



VRAAG/QUESTION 3 [24]

Vraag 3.1 [2]

Gebruik fasor-analise en bepaal $v(t)$:

$$v(t) = -20\cos(\omega t - 60^\circ) + 30\sin(\omega t - 30^\circ) \text{ V}$$

Question 3.1 [2]

Use phasor analysis to find $v(t)$:

$$v(t) = -20\cos(\omega t - 60^\circ) + 30\sin(\omega t - 30^\circ) \text{ V}$$

Vraag 3.2 [2]

Bepaal $\bar{I}_2$ in polêre form:

$$\begin{bmatrix} 5 + j3 & -j3 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} \bar{I}_1 \\ \bar{I}_2 \end{bmatrix} = \begin{bmatrix} 5\angle 0^\circ \\ 0 \end{bmatrix}$$

Question 3.2 [2]

Determine $\bar{I}_2$ in polar form:

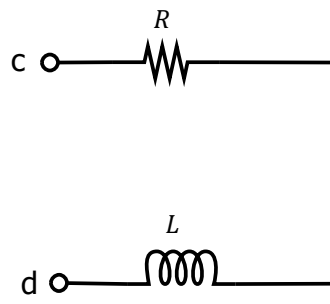
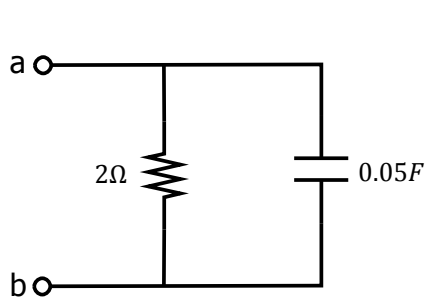
$$\begin{bmatrix} 5 + j3 & -j3 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} \bar{I}_1 \\ \bar{I}_2 \end{bmatrix} = \begin{bmatrix} 5\angle 0^\circ \\ 0 \end{bmatrix}$$

Vraag 3.3 [4]

Bepaal R en L as die intree-impedansies van die twee stroombane presies dieselfde moet wees by $\omega = 10 \text{ rad/s}$.

Question 3.3 [4]

Determine R and L if the input impedances of the two circuits have to be exactly the same at $\omega = 10 \text{ rad/s}$.



Vraag 3.4 [4]

Gebruik node-analise om twee vergelykings van die volgende vorm op te stel:

$$(a_1 + jb_1)\bar{V}_1 + (a_2 + jb_2)\bar{V}_2 = 20\angle 90^\circ$$

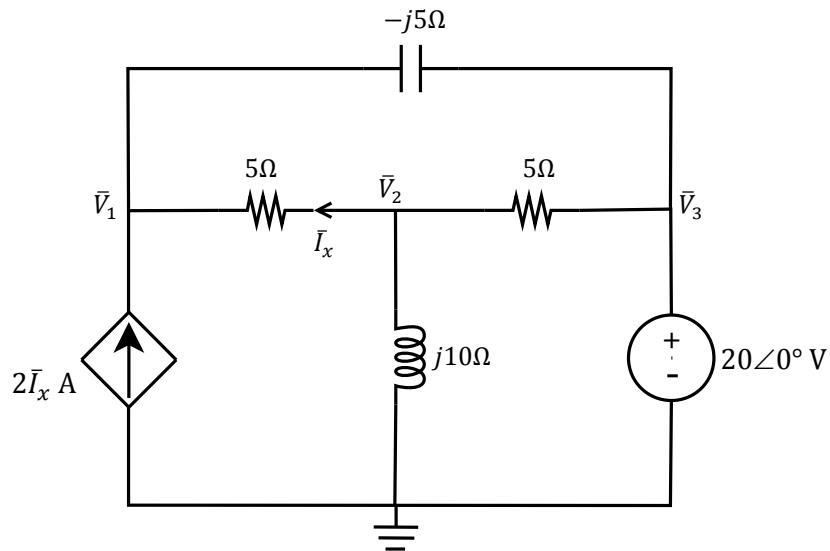
$$(a_3 + jb_3)\bar{V}_1 + (a_4 + jb_4)\bar{V}_2 = 40\angle 0^\circ$$

Question 3.4 [4]

Use nodal analysis to set up two equations of the following form:

$$(a_1 + jb_1)\bar{V}_1 + (a_2 + jb_2)\bar{V}_2 = 20\angle 90^\circ$$

$$(a_3 + jb_3)\bar{V}_1 + (a_4 + jb_4)\bar{V}_2 = 40\angle 0^\circ$$

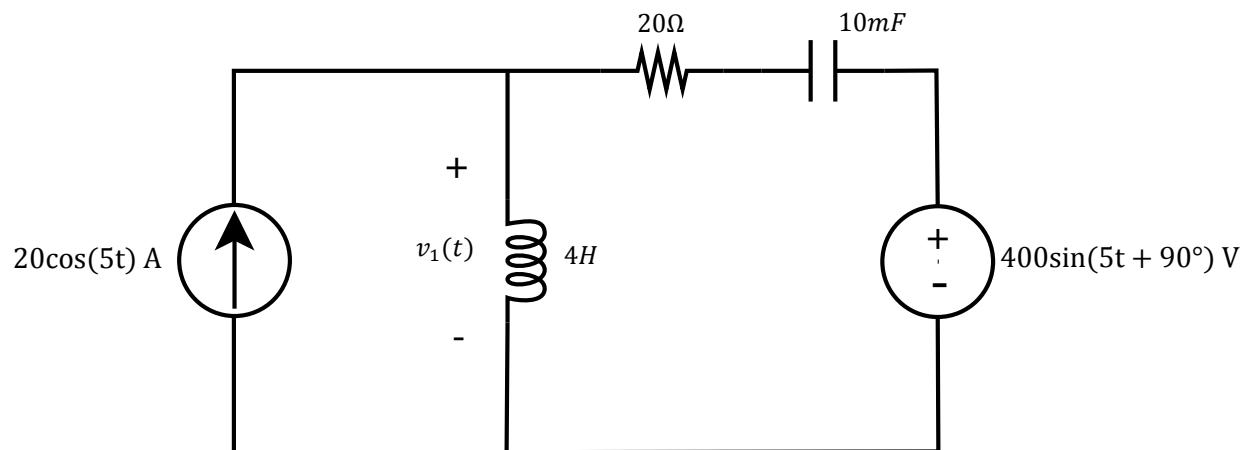


Vraag 3.5 [4]

- a) Bepaal $v_1(t)$ slegs agv die stroombron.
b) Bepaal $v_1(t)$ slegs agv die spanningsbron.

Question 3.5 [4]

- a) Determine $v_1(t)$ only due to the current source.
b) Determine $v_1(t)$ only due to the voltage source.

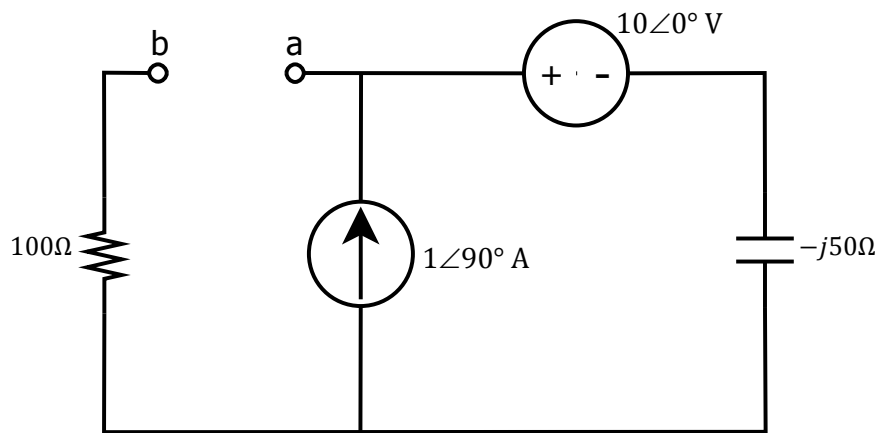


Vraag 3.6 [4]

Bepaal die Norton-ekwivalent tov terminale a-b. Skryf die antwoorde as fasors in reghoekige vorm.

Question 3.6 [4]

Find the Norton equivalent wrt terminals a-b. Write the answers as phasors in rectangular form.

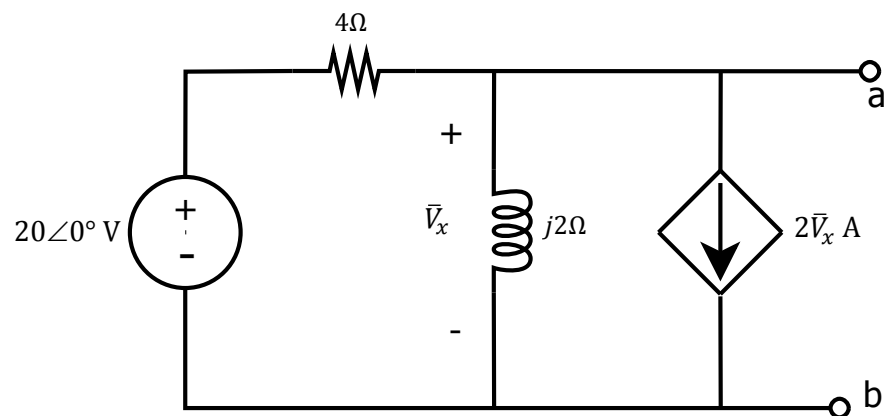


Vraag 3.7 [4]

Bepaal die Thevenin-ekwivalent tov terminale a-b. Skryf die antwoorde as fasors in polêre vorm.

Question 3.7 [4]

Find the Thevenin equivalent wrt terminals a-b. Write the answer as phasors in polar form.



EBN122 EKSAMEN/EXAMINATION (23 NOVEMBER 2015)
ANTWOORDBLAD / ANSWER SHEET

Van / Surname:	MEMO	Studentennummer: Student number:							
Voorletters / Initials:		Tel. nr. / Tel no.:							

Q1		Q2		Q3		Tot:	/90
----	--	----	--	----	--	------	-----

VRAAG 1 - QUESTION 1 (45)	
1.1	$n = 125 \times 10^6$ ✓
1.2	$i_{(0.2s)} = 4A$ ✓✓
1.3	$W = 324 kJ$ ✓✓
1.4	$Cost = R 194.4$ ✓✓
1.5	a) $P_{(I)} = -326.3$ ✓ b) $P_{(V)} = -86.52 W$ ✓✓
1.6	a) Nodes = 5 ✓ b) Branches = 7 ✓
1.7	a) $R_{eq} = 6.29 \Omega$ ✓✓ b) $R_{eq} = 6 \Omega$ ✓✓
1.8	$V_o = -44.1 V$ ✓✓ $I_o = -2.94 A$ ✓✓
1.9	$a = 16$ ✓ $b = 0$ ✓ $c = -20$ ✓ $d = 10$ ✓
1.10	$e = 4$ ✓ $f = 2$ ✓ $g = 1$ ✓ $h = -5$ ✓
1.11	a) $I_B = 28.7 mA$ ✓✓ b) $V_{CE} = 10V$ ✓✓
1.12	a) $V_{I(2A)} = -180V$ ✓✓ b) $V_{I(5V)} = 50V$ ✓✓
1.13	a) $I_N = 3A$ ✓✓ b) $R_L = 5 \Omega$ ✓✓
1.14	$V_{th} = \frac{-55}{19}$ or $\frac{-10}{19}$ $R_{th} = 3/7$ or 0.4286Ω ✓✓ -3.93

VRAAG 2 - QUESTION 2 (22)		
2.1	a) $v_{01} = -1V$ ✓	b) $v_{01} = 6V$ ✓
2.2	a) $v_{02} = 10V$ ✓	b) $v_s = 4V$ ✓
2.3	a) $v_{03} = 0V$ ✓	b) $i_{03} = -0.1mA$ ✓
2.4	$i_1(2s) = 24A$ ✓	$i_2(4s) = 2.2mA$ ✓
2.5	$L_{eq} = 16mF$ ✓	
2.6	$W_L = 2J$ ✓	$W_C = 400J$ ✓

VRAAG 3 - QUESTION 3 (24)		
3.1	$v(t) = 26.46 \cos(\omega t - 160.9^\circ)$ ✓	
3.2	$\bar{I}_2 = 1.07 \angle 115.5^\circ$ ✓ $(-0.46 + j0.97)$	
3.3	$R = 1\Omega$ ✓	$L = -0.1H$ ✓
3.4	$a_1 + jb_1 = 3 + j$ ✓	$a_2 + jb_2 = -3$ ✓
	$a_3 + jb_3 = -2$ ✓	$a_4 + jb_4 = 4 - j$ ✓
3.5	$v_1(t)_{(current\ source)} = 565.7 \cos(5t + 45^\circ) V$ ✓	
	$v_1(t)_{(voltage\ source)} = 400 \cos(5t + 90^\circ) V$ ✓	
3.6	$\bar{I}_N = 0.54 \angle 26.6^\circ$ $0.48 + j0.24 A$ ✓	$\bar{Z}_N = 111.8 \angle -26.6^\circ$ $100 - j50\Omega$ ✓
3.7	$\bar{V}_{Th} = 2.17 \angle 12.5^\circ V$ ✓ $(2.11 + j0.47)$	$\bar{Z}_{Th} = 0.43 \angle 12.5^\circ$ ✓ $(0.42 + j0.09)$